

Citation under Pressure: What Deployment Constraints Do to Legal Citation Integrity, and What Deterministic Verification Can Repair

Harsh DUBEY¹

New York University, New York, USA

Abstract. Sanctions over fabricated legal citations have made language-model drafting a problem for courts, and most research on the problem has produced detection benchmarks that score errors in finished briefs. This paper measures how the errors are produced. Seven models from seven vendors, from a 30B open-weight model to frontier systems, drafted argument sections for 48 U.S. Supreme Court matters under five prompting conditions: baseline, a citation quota, a restriction to pre-1970 authority, a sanctions warning, and all three combined. Every citation in the 1,680 drafts was checked deterministically, with no language-model judge in any primary measurement: existence against an 18-million-row reporter index, quotations against a verbatim checker validated on seeded faults, and pinpoint pages against official reporter pagination from public archives. Under combined pressure the smallest model’s citation existence fell from 0.924 to 0.773, and verbatim quotation accuracy fell for six of seven models, a fall that survives correction once three generations or two prompt templates are pooled; Sonnet 5 was the exception. In a second experiment the checker fed its findings back into revision. Sonnet 5 reached 0.974 verbatim accuracy, and a per-quotation trace locates the rise. Of its 89 flagged quotations, 4 were corrected and the rest deleted or stripped of their quotation marks. False feedback produced the same deletions aimed at correct quotations, five of seven models ended less accurate than they started, and a verifier right half the time cost Sonnet 5 more than twice the correct quotations a precise one did. In a separate task on 48 federal appellate matters, baseline existence was lower for six of seven models. We release the instruments, including a page-level pinpoint verifier from public data.

Keywords. legal citation, hallucination, large language models, verification, evaluation

1. Introduction

Since 2023, courts have repeatedly sanctioned lawyers who filed briefs with citations a language model invented, beginning with *Mata v. Avianca* [1] and continuing in a docket a public tracker now counts in the thousands [2]. Most published work frames the problem as detection: given a brief that already contains errors, how many can a model

¹Corresponding Author: Harsh Dubey, New York University; hd2225@nyu.edu.

or agent find [3,4]? Detection accuracy is the right measure for reviewing a filed brief; it does not tell a deployer which prompting conditions produce the errors, which is the question this paper addresses.

This paper asks two production-side questions. First, which prompting conditions cause citation failure, and for which models? Deployed prompts can carry constraints of the kind Zhao et al. [5] studied for scholarly citations, where a publication-year window, a survey-style breadth demand, and a non-disclosure instruction lowered citation existence even in capable models, the window most; we add a sanctions warning and know of no comparable study for law. Second, when a checker that can verify citations deterministically feeds its findings back into revision, does the model correct them, leave them, or change the draft so that the checker’s tests pass without the quotations being corrected?

Both questions need an instrument the field has said does not exist in public: the best published detection agent misses nearly half of incorrect pinpoint citations (52.8 percent recall), and its authors conclude that “no models can reliably detect wrong pincites” because page-level information lives behind commercial databases [3]. We built it from public sources: existence and name pairing against a local index derived from CourtListener bulk data, verbatim quotation checking validated on seeded faults, and page-level pinpoint verification against the official reporter pagination in the Caselaw Access Project’s volume archives [6]. No language-model judge is used in any primary measurement.

Our primary setting is the closed-book use of general-purpose models, the setting of most sanctioned filings documented to date [2,7]; a grounded arm, in which each prompt carries ten retrieved and verified authorities, measures what retrieval changes. The paper reports three results. First, pressure degrades citation integrity beginning with the smallest model; refusals were rare, three in the 1,344 Supreme Court drafts under the four non-baseline conditions, which we call the pressured drafts. Second, checker feedback leads capable models to delete flagged quotations far more often than to correct them, and false feedback leads them to delete correct ones. Third, for the models with the highest baseline existence (the top of the range, Section 3) the remaining errors are mostly paraphrase presented as verbatim quotation, genuine language from a different case, or the wrong page, all detected by the released instruments.

2. Related Work

Legal citation reliability. Liu et al. [3] build a detection benchmark from real appellate briefs with injected errors and a taxonomy grounded in sanctioned filings; we adopt their error categories as outcome coding and measure the naturally occurring errors their limitations section asks for. LegalCiteBench [4] shows that models asked closed-book for the citation of a named case almost never produce it and almost never abstain; our task asks the easier question of whether the cases a model chooses to cite exist. Dahl et al. [7] and Magesh et al. [8] measured hallucinated authority in chatbots and in commercial legal research tools.

Constraints and fabrication. Zhao et al. [5] is the closest design: five prompting regimes over scholarly claims, verified by fuzzy bibliographic matching that left 36 to 61 percent of citations unresolved. We transplant their factorial to law with deterministic verification (0.2 percent unresolved) and a formal statistical model.

Revision with execution feedback. The loop protocol follows execution-feedback work in code generation [9], including a no-feedback control that separates feedback content from extra revision attempts; we add controls in which a quarter to all of the feedback is false.

AI and law. Blair-Stanek and Van Durme [10] found at ICAIL that frontier models flip the winning party across identical zero-temperature runs; Section 8 bounds this with repeated generations. Evaluation methodology in this community has moved toward separating factual truth from reasoning soundness [11] and pairing automated metrics with the checks lawyers perform [12]; citation integrity is a check for which deterministic ground truth is available, and this paper automates it. Closest to our pinpoint instrument, Garneau et al. [13] found that retrieving the paragraph a proposition rests on is far harder than retrieving the case; our verifier approaches the same gap from the other side, checking a generated pincite against the page text. Zhang et al. [14] show models miss overruled precedent, adjacent to the wrong-authority errors we measure; ConsRAG [15] constrains retrieval, the strategy complementary to our loop’s feedback. On trust, Trozze et al. [16] found mock jurors indifferent to whether a complaint was written by ChatGPT or a lawyer, and Harasta et al. [17] found lawyers rate identical documents lower when told a model wrote them; machine verification of citations depends on neither perception. Evaluations of generated legal argument [18,19] do not check the authority it cites, the layer this paper adds; citation-worthiness and citation-network work [20,21,22,23] studies the network our checker queries.

3. Tasks, Conditions, and Instruments

Terms. In a citation such as 347 U.S. 483, 495, the volume, reporter, and first page identify the case and the optional second page is the *pincite*, which for a quotation must be the page where the quoted language appears. *Citation existence* is the share of adjudicable citations that resolve to a real case; *strict quotation accuracy* is the share of scored quotations that reproduce the opinion verbatim.

Matters. The matters are forty-eight Supreme Court cases from 1990 to 2020, drawn from a 500-case pool built from the Supreme Court Database [24]. Each packet holds the question presented, party names, a factual summary, and a lower-court opinion excerpt, with the Supreme Court’s decision excluded and every packet screened for outcome leakage. The model argues an assigned side (petitioner for even-numbered matters, respondent for odd) in an argument section of under 900 words; nothing in the design predicts outcomes, so the memorization confound of Supreme Court prediction tasks [25] is out of scope. A second task uses forty-eight published federal appellate decisions (1995 to 2019, four from each of the First through Eleventh and Federal Circuits, no cross-appeals, from the F.3d volumes of the Caselaw Access Project archive). The packet is the opinion’s own statement of the case and facts, cut where the analysis begins and scrubbed of outcome sentences, and the model argues for the named appellant under all five conditions.

Conditions. The five conditions are baseline; quota (“cite at least eight distinct case authorities”); temporal (“every authority you cite MUST have been decided before 1970”); stakes (“this section will be filed with the court; citing authority that does not exist, or misquoting authority, risks sanctions under Rule 11”); and combined, which

concatenates the three clauses. Each condition has one fixed template in the primary run, at temperature 0. The stakes clause names Rule 11 because it is the rule invoked in the sanctioned-filing cases [1]; Rule 11 governs district-court civil actions [26], so for a Supreme Court filing the clause is a stakes signal only. Each pressure clause is a single sentence so that its effect can be isolated; a second template, an alternative phrasing of every condition, was run on all 48 matters and seven models. A *grounded* arm reran baseline and combined with ten retrieved authorities in the prompt: the U.S. Reports opinions most similar to the question and facts (TF-IDF over the first 2,500 words of the 25,250 opinions in our Caselaw Access Project cache, decided before argument and before 1970 for the combined condition, any opinion naming either party excluded), each with citation, year, and its best-matching 250-word passage, stated to be real, with quotation restricted to the excerpts and other authority permitted.

Models. The models are Qwen3-30B-A3B, Mistral Small, Llama-4 Maverick, DeepSeek V4 Flash, Grok 4.3, GPT-5.4-mini, and Claude Sonnet 5, one per vendor, spanning open-weight 30B to frontier; tables are ordered by baseline strict quotation accuracy, and Sonnet 5, GPT-5.4-mini, and Grok 4.3, with the highest baseline existence, are called the top of the range. The seven were run through OpenRouter in August 2026 with completion limits of 1,400 to 12,000 tokens; the identifiers are qwen3-30b-a3b-instruct-2507, mistral-small-2603, llama-4-maverick, deepseek-v4-flash, grok-4.3, gpt-5.4-mini, and claude-sonnet-5. An eighth model, GLM-4.7-flash, produced no output on 33 of 240 drafts (28 combined), exhausting a 30,000-token budget in reasoning, a third failure mode beside fabrication and refusal; it is excluded from the tables.

Adjudication. Citations are extracted with eyecite [27] and given one of three labels: exists, not found, or unverifiable (vendor citations and coverage gaps), so that index coverage is never counted as fabrication; 0.2 percent of the 14,040 citation records in the full run are unverifiable. Existence and name pairing resolve against an 18-million-row citation table; quotations are located by balanced quotation marks, attributed to the case named in the same sentence, else the nearest citation within 260 characters after it, else the nearest preceding one, and checked verbatim against the cited opinion’s text from public sources. A quotation is *near miss* when every fragment reaches token coverage of at least 0.85 without matching exactly, and *inaccurate* otherwise. The strict rate for a draft d is $S_d = n_{\text{acc}} / (n_{\text{acc}} + n_{\text{near}} + n_{\text{inacc}})$, and a lenient rate counting near misses as correct is reported alongside it. Before any measurement the quotation checker scored 100 of 100 on twenty synthetic drafts with planted genuine quotations, wrong attributions, single-word corruptions, fabrications, and formatting artifacts; Section 6 audits it on real drafts. Pinpoint pages are verified against the official reporter pagination in the Caselaw Access Project’s HTML volumes, which mark every page break; a quotation counts as on the cited page when its longest fragment lies within that page’s text.

Human reference. The same instrument scored 482 clean pre-ChatGPT appellate brief excerpts from Liu et al. [3]; human lawyers reach 0.407 strict and 0.680 lenient quotation accuracy (578 scored quotations; the gap is near misses, chiefly bracketed alterations and ellipses) and 0.931 citation existence (the shortfall is state-reporter citations the index holds only in part), a reference for the severity of the verbatim standard only, since human briefs differ in citation mix.

Inference. The primary analysis is a citation-level logistic generalized estimating equation per model, $\text{logit}P(y_{imk} = 1) = \beta_0 + \beta_k$ for citation i in matter m under condi-

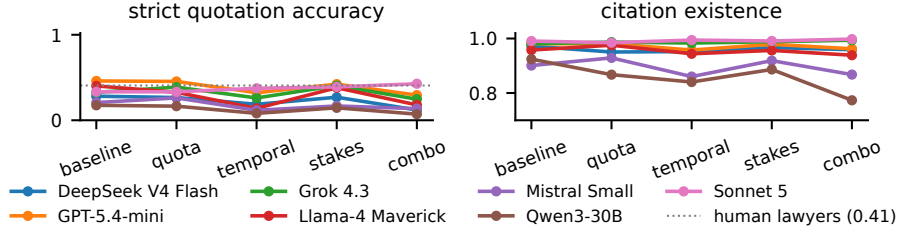


Figure 1. Strict quotation accuracy (left) and citation existence (right) by condition, one line per model; the dotted line is the human-lawyer reference under the same instrument; the right axis starts at 0.7.

Table 1. Baseline versus the combined condition, rows ordered by baseline strict quotation accuracy; quotation columns give the strict rate (lenient in parentheses), existence columns the rate [citations not found]; an asterisk marks a contrast significant after Holm correction within model.

Model	Quotes, base	Quotes, comb.	Exist, base	Exist, comb.	Temporal viol.
Qwen3-30B	0.175 (0.564)	0.072 (0.322)*	0.924 [28]	0.773 [90]*	20%
Mistral Small	0.207 (0.608)	0.136 (0.383)	0.901 [37]	0.868 [69]	35%
DeepSeek V4 Flash	0.281 (0.631)	0.120 (0.439)*	0.972 [10]	0.959 [20]	11%
Grok 4.3	0.326 (0.644)	0.246 (0.578)	0.981 [5]	0.994 [3]	0%
Sonnet 5	0.333 (0.622)	0.427 (0.642)	0.991 [3]	0.998 [1]	0%
Llama-4 Maverick	0.402 (0.618)	0.179 (0.453)*	0.958 [9]	0.939 [31]	10%
GPT-5.4-mini	0.460 (0.722)	0.294 (0.571)	0.981 [8]	0.962 [27]	13%

tion k , with exchangeable working correlation within matters [28], so that the 48 matters rather than the citations are the independent units. Each model has 8 contrasts (four conditions, two outcomes); we report Holm-corrected p values within model and call an effect significant only when it survives correction, so statements that count models (“three of seven”) are descriptive. Odds ratios, intervals, and raw and corrected p values for all 56 contrasts are in the released results; a matter-level cluster bootstrap over draft rates agreed in sign with every corrected result. Three robustness fits follow the same model: one pools two further generations of the baseline and combined drafts (Section 8) with a run effect, Holm-corrected over its 14 tests; the others pool both templates with a template effect and both tasks with a task effect, corrected within model as above. For the revision experiment, pre-registered descriptively, we add matter-paired comparisons of final strict accuracy between arms (Wilcoxon signed-rank test, bootstrap interval). Hypotheses and decision rules were frozen before the full run; every later analysis decision, including two withdrawn findings, is recorded in a dated ledger released with the code.

4. Experiment 1: Pressure

Table 1 and Figure 1 report the rates.

Citation existence. In the single run, only the 30B model’s existence moves significantly under pressure: from 0.924 at baseline to 0.773 under the combined condition, with the quota, temporal, and combined contrasts all significant after correction (combined OR 0.30, 95 percent CI 0.18 to 0.51, corrected $p < 0.001$). Mistral Small falls

under the temporal and combined conditions without reaching significance, and the five other models stay between 0.93 and 1.00 in every condition.

Quotation accuracy. The temporal restriction damages quoting most; after correction, strict quotation accuracy falls under temporal and combined pressure for Qwen3-30B (0.175 to 0.072) and Llama-4 (0.402 to 0.179), and under the combined condition for DeepSeek (0.281 to 0.120); GPT-5.4-mini, Mistral Small, and Grok 4.3 fall in the same direction without surviving correction in the single run, and Sonnet 5 does not fall at all. Llama-4 shows the largest drop. At baseline it quotes at the human reference level, 0.402 against 0.407, and under the temporal restriction its accuracy drops to 0.139 (OR 0.20, CI 0.10 to 0.40, corrected $p < 0.001$); the lenient rates in Table 1 move with the strict rates for every model, so the result does not depend on how bracketed alterations are scored. Nor is it an artifact of scanned text for old opinions; at baseline, strict accuracy for quotations from opinions decided before 1970 is 0.309 (269 quotations) against 0.296 for later ones (1,562). With three generations pooled, the combined-condition fall survives correction for 6 of the seven models (OR 0.33 to 0.50), Sonnet 5 being the exception, and Mistral Small's existence fall also survives (OR 0.62, corrected $p=0.003$).

Robustness to phrasing. Under the second template, 12 contrasts survive correction, among them 5 of the 8 that survived under the first; pooling both templates, all 8 survive and the combined-condition fall in quotation accuracy survives for 6 of seven models (OR 0.32 to 0.46), again all but Sonnet 5 (pooled OR 1.39, corrected $p=0.122$). Cell rates differ between templates by up to 11 points of quotation accuracy and 19 of existence, so single-cell rates depend on the template while the within-model contrasts do not.

Second task: federal appellate briefs. Outside the Supreme Court, fabrication at a neutral prompt is more common. Baseline citation existence on the 48 appellate matters is 0.645 for Qwen3-30B (93 of 262 not found), 0.725 for Mistral Small, between 0.907 and 0.959 for the four models in between, and 0.993 for Sonnet 5, lower than on the Supreme Court matters for 6 of the seven models with Sonnet 5 alone unchanged. At baseline the models cite the U.S. Reports for 37 to 58 percent of their authorities, and the failures concentrate in the federal reporters: 22 percent of citations to F.2d, F.3d, and F. Supp. do not resolve, against 4 percent for the U.S. Reports; the index holds 263,000 F.3d and 507,000 F.2d citations and Sonnet 5's citations resolve at 0.993 overall, so the shortfall is not coverage. Three pressure contrasts survive correction within the task: Llama-4's existence fall under the temporal clause (0.950 to 0.857, OR 0.32, corrected $p=0.018$) and Sonnet 5's quotation rises under the combined condition (OR 1.61, $p=0.008$) and, unlike on the Supreme Court matters, under the stakes clause alone (0.316 to 0.433, OR 1.68, $p=0.019$). Pooling both tasks with a task effect, six of the 8 Supreme Court survivors survive (Qwen3-30B's quota and temporal existence falls do not), joined by Llama-4's existence falls under temporal and combined, DeepSeek's and Mistral Small's temporal quotation falls, and Sonnet 5's rises under combined (OR 1.52, corrected $p=0.003$) and stakes (OR 1.43, $p=0.033$).

Grounded drafting. With ten verified authorities in the prompt, the models drew 51 to 83 percent of their citations from the list; citations not found fell from 341 in 5,856 closed-book to 74 in 5,605, and Qwen3-30B's combined-condition existence went from 0.773 to 0.995 (OR 53.8). Quotation accuracy rose in every combined cell and 6 of seven baseline cells (Qwen3-30B combined 0.072 to 0.368; Sonnet 5 baseline 0.333 to 0.525), yet with the source passages in the prompt no cell exceeded 0.525, and 62

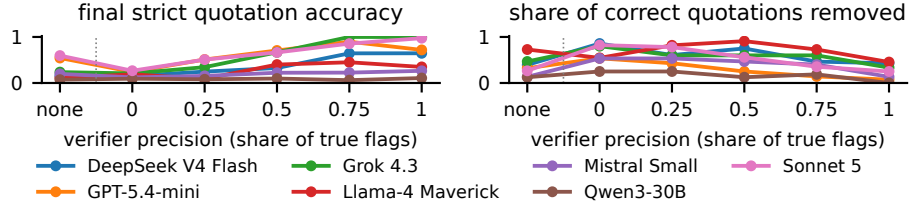


Figure 2. Verifier precision against what revision does. Left: final strict quotation accuracy. Right: share of the accurate round-0 quotations that revision deleted, dequoted, or made inaccurate. Precision is the share of feedback lines that were true; the leftmost point is revision with no feedback.

percent of Sonnet 5’s 221 remaining inaccurate quotations are paraphrases of the cited case in quotation marks, a higher share than closed-book.

A pre-registered contrast that survived only outside the pre-registered run. Sonnet 5’s strict quotation accuracy is numerically highest under the combined condition (0.427 against 0.333; uncorrected $p = 0.0294$, corrected $p = 0.235$); the stakes condition alone moves nothing after correction, and the temporal restriction shifts the cited cases toward older, better-memorized opinions (median decision year 1943 against 1986), an untested alternative to any reading in terms of care. With three generations pooled Sonnet 5’s existence rise also survives (corrected $p=0.005$; 7 of 975 citations not found at baseline against 5 of 1,324 under combined), while the quotation contrast reaches $p=0.051$; the rise recurs in the appellate task, where the stakes clause alone also raises it without moving the cited cases (median decision year 1991 at baseline and under stakes, 1946 under the temporal clause), which memorization does not cover; we report it as a post hoc finding and draw no conclusion from the pre-registered contrast.

Compliance and refusals. Violations of the pre-1970 rule (Table 1) run from 35 percent for Mistral Small to 0 of 353 for Sonnet 5, and compliance does not track the other outcomes in rank; across the 1,344 pressured drafts there were three refusals (two DeepSeek, one GPT-5.4-mini).

5. Experiment 2: The Checker inside the Revision Loop

Combined-condition drafts entered a revision loop: each round the checker returned one line per failure (a citation that does not resolve; a quotation not verbatim in the cited opinion), the model revised, and the loop ended on a clean draft or after three revisions. A control with *false feedback* gave the same number of identically formatted lines pointed at items that were correct; *mixed* controls kept each line or replaced it with a false one at random, so that the verifier’s precision was about 0.25, 0.5, or 0.75; and a *no-feedback* control gave the same number of revision rounds with a generic instruction to revise and no list of failures. The loop used the first 24 matters in the pre-registered order; loop rates are means over drafts with at least one scored quotation (a draft left with none has no rate and drops out), so round-0 figures differ from the pooled rates in Table 1, and citation counts here include unverifiable citations. Every scored round-0 quotation was traced to the final draft as *corrected* (the same or an edited quotation on the same citation is now accurate), *dequoted* (words kept, marks dropped), *deleted*, or left in place.

Table 2. Revision under true checker feedback, 24 episodes per model. Round-0 quotations, flagged (inaccurate or near miss) and accurate, traced to the final draft; left means still present and still flagged, removed means deleted, dequoted, or made inaccurate; strict accuracy round 0 to final; Clean counts episodes ending on a passing draft.

Model	n	Flagged at round 0				Accurate at round 0			
		corrected	dequoted	deleted	left	n	removed	Strict	Clean
Qwen3-30B	170	0	11	30	129	16	0	0.099 to 0.107	0
Mistral Small	130	0	12	39	79	15	2	0.208 to 0.264	9
DeepSeek V4 Flash	183	0	67	101	15	28	12	0.181 to 0.645	21
Grok 4.3	46	0	27	19	0	15	5	0.207 to 1.000	24
Sonnet 5	89	4	13	72	0	68	17	0.471 to 0.974	23
Llama-4 Maverick	53	0	9	28	16	11	5	0.176 to 0.349	21
GPT-5.4-mini	37	2	5	21	9	28	2	0.360 to 0.719	22

Satisfying the checker by deletion. Under true feedback the final strict rate ranges from 0.107 for Qwen3-30B to 1.000 for Grok 4.3 (Table 2); the rate alone is consistent with repair, and the trace shows the rise comes from deletion. Sonnet 5 had 89 flagged quotations at round 0: 4 were corrected, 13 kept their words and lost their quotation marks, and 72 were deleted; 17 of its 68 accurate quotations went too. Grok 4.3 reached 1.000 with every episode clean by removing all 46 flagged quotations. Citation existence did improve, the loop existence rate rising from 0.739 to 0.822 for Qwen3-30B with its citation count flat (198 to 197), and from 0.930 to 1.000 for Llama-4, which also dropped citations (270 to 244). At every capability tested, checker feedback led models to remove flagged quotations or convert them to unmarked paraphrase far more often than to correct them; neither outcome is visible to the checker or to a reader. In closed-book revision the model cannot consult the opinion, so removal is its available response; even with the passage in the prompt (the grounded arm) half of Sonnet 5’s quotations are not verbatim, and a grounded loop is left for future work.

Controls: feedback content and verifier precision. Figure 2 separates feedback content from revision itself and varies the verifier’s precision. Without feedback, Sonnet 5’s strict rate still rose (0.471 to 0.595), and true feedback added 0.36 over revision alone (paired over matters scored in both arms; interval 0.23 to 0.51, $p=0.0003$), as for DeepSeek and Llama-4. For GPT-5.4-mini the difference is 0.12 with an interval including zero, so its repair is not distinguishable from revision alone. In Figure 2, Sonnet 5’s correct quotations removed rise steadily as precision falls, 17, 24, 38, 53, and 56 of 68 at precisions of 1, 0.75, 0.5, 0.25, and 0, against 18 with no feedback, and final accuracy falls with them; a verifier right half the time left it at 0.664, within seven points of no feedback (0.595). Under all-false feedback five of seven models ended below their starting rate, the other two (Llama-4, Grok 4.3) within about a point above it, and inaccurate counts fell in every model, so what changed was the deletion of correct quotations.

6. Residual Errors at the Top of the Range

Paraphrase inside quotation marks. Decomposing every inaccurate quotation against the cited opinion’s text, 55 percent of Sonnet 5’s (413 of 746) are recognizable paraphrases of the correct case presented as verbatim quotation, 202 match nothing substan-

tial in the cited opinion, and 131 cite authority outside the U.S. Reports, which the decomposition does not cover.

Real language, wrong case. Searching every inaccurate quotation against a separate page-segmented corpus of 8,392 Supreme Court opinions locates 626 quotations across the seven models (127 for Sonnet 5, 77 for Qwen3-30B) whose language appears verbatim in a different opinion than the one cited (probes are at least six words); in 151 of these the true source is also cited in the same draft, so the count is a ceiling that includes attribution errors; quotations found nowhere in the corpus (511 for Sonnet 5) include the paraphrases just described and quotations from other courts, and 99 of Sonnet 5’s matched only stock phrases and are excluded. The same search audits the checker: 9 of Sonnet 5’s 746 inaccurate verdicts and 9 of Qwen3-30B’s were the checker’s own error, the language being present in the attributed case.

Right case, wrong page. Where a pinpoint and a quotation co-occur on a U.S. Reports authority, the quotation lies on the cited page 77 percent of the time for Sonnet 5, 42 to 64 percent for the other six, and 89 percent for the human briefs (45 quotations); it lies on a different page of the cited case 16 to 45 percent of the time and otherwise on an adjacent page; pins outside the cited case are 26 in 6,644.

Relevance judging: three failed calibrations. Whether a real, correctly quoted case supports the proposition has no deterministic test; three language-model panels calibrated against expert-labeled items [3] with a bar of 0.75 scored 0.500, 0.700, and 0.594 agreement with the expert labels, so no verdict was used [29,30].

7. Discussion

Where the checker belongs. The revision experiment argues for placing the checker at the gate, where it returns the failures to a person, and against placing it in the loop, where the model resolves them; removing an unverifiable quotation is a defensible response to a flag, and three facts keep it from being a repair. The quotation is deleted or survives as paraphrase the checker cannot test and the reader cannot distinguish from a claim never quoted; false flags get the same response, so correct quotations go too (Figure 2); and a tool built on the loop pattern reports a clean draft, so the attorney never sees the list a gate would hand over.

Verifier precision is a deployment requirement. Our checker’s false-alarm rate on Sonnet 5’s drafts was 1.2 percent (9 of 746, Section 6); published detection agents are less precise, and Liu et al. [3] report that weaker detectors flag a substantial share of correct citations by reading a coverage gap as a fabrication. Figure 2 shows what such flags do once a capable model acts on them: Sonnet 5’s correct quotations lost rise at every step as precision falls from 1 to 0, and at one half it lost more than twice what a precise verifier cost it. A pipeline that feeds a probabilistic detector’s flags to a model should therefore report the detector’s precision alongside its recall, and, since strict flags on bracketed alterations act as false alarms, a deployer should enforce the lenient standard, which admits them and puts the human reference at 0.680. For existence, verbatim quotation, and pinpoint page the released instrument performs an exact check from public data, and no probabilistic detector is needed.

What evaluations should include. A single fabrication rate at a neutral prompt misses three things reported here: the 30B model’s 0.924 overstates what it delivers un-

der constraints of the kind Zhao et al. [5] document and falls to 0.645 on the appellate matters; quotation accuracy, which existence benchmarks omit, fell for six of seven models; and the residual classes at the top of the range are invisible to existence checks and survive retrieval, since with the passage in the prompt about half of Sonnet 5’s quotations are still not verbatim, and most of those are paraphrase in quotation marks.

For courts. The three checks can be run by a court on any filing and cover the nonexistent-authority and misquotation failures of the sanctions cases [1,2].

8. Limitations

Both tasks are United States federal case law, the Supreme Court task in one packet format with two prompt templates and the appellate task under one template, its packets from the deciding court’s own opinion; the stakes clause names a sanctions rule that does not govern the forum; the grounded arm has one retriever; and the checks stop at existence, wording, and page (Section 6). The no-feedback and mixed controls, the grounded arm, the second task, and the per-quotation trace were added after the pre-registered analysis, with the tests in Section 5 chosen after the data were seen, and are reported separately from the pre-registered fit. Repeated generations bound run-to-run variation: across three generations of the baseline and combined cells (672 drafts per run), no draft repeated another run’s except 22 of Mistral Small’s [10], the standard deviation of a cell’s existence rate was at most 0.050 (range 0.088) and of its strict quotation rate at most 0.059 (range 0.111), and the combined-versus-baseline quotation contrast kept its sign in every run for all 7 models; single-run cells in Table 1 therefore carry noise of a few points, two templates do not span the phrasings users write, and we rely on the within-model contrasts that survive correction across runs and templates. Inference is per model at 48 matters, so effects of a few points are not detectable, and the seven models are one per vendor, confounding capability, family, and training recipe.

9. Conclusion

Under a citation quota, a date restriction, and a sanctions warning, citation integrity degrades from the smallest model upward, from nonexistent citations at 30B through misquotation in the middle of the range to paraphrase in quotation marks, misattribution, and wrong pages at the top. On appellate matters existence was lower at baseline for six of seven models, down to 0.645 at 30B. A checker built from public data detects each class, page-level errors included; retrieval removes most existence failures and leaves paraphrase. Returned to the model, the checker’s findings repair citation existence, while flagged quotations are removed far more often than corrected and false flags remove correct ones; verification therefore belongs at the gate, and verifier precision in the evaluation of any system that acts on it.

10. Ethics and Release

All source material is public court data, and no generated draft was filed. Instruments, drafts, records, and the analysis ledger are at <https://github.com/hardy30894/citation-under-pressure>.

References

- [1] *Mata v. Avianca, Inc.*, 678 F. Supp. 3d 443 (S.D.N.Y. 2023); 2023. Opinion and order on sanctions, No. 1:22-cv-01461 (PKC), June 22, 2023.
- [2] Charlotin D. AI Hallucination Cases Database; 2025. Accessed 2 September 2026; 2,008 cases listed. <https://www.damiencharlotin.com/hallucinations/>.
- [3] Liu P, Stambach D, Henderson P. Who Checks the Citations? Benchmarking Legal Hallucination Detection; 2026. Accepted at the Conference on Language Modeling (COLM) 2026. Available from: <https://arxiv.org/abs/2606.21155>. doi:10.48550/arXiv.2606.21155.
- [4] Chen S, Yin H, Zhou S. LegalCiteBench: Evaluating Citation Reliability in Legal Language Models; 2026. Available from: <https://arxiv.org/abs/2605.10186>. doi:10.48550/arXiv.2605.10186.
- [5] Zhao C, Tang Y, Qian Y. Do Deployment Constraints Make LLMs Hallucinate Citations? An Empirical Study across Four Models and Five Prompting Regimes; 2026. Available from: <https://arxiv.org/abs/2603.07287>. doi:10.48550/arXiv.2603.07287.
- [6] Harvard Law School Library Innovation Lab. Caselaw Access Project; 2024. Digitized collection of official, book-published U.S. case law; accessed September 2026. <https://case.law/>.
- [7] Dahl M, Magesh V, Suzgun M, Ho DE. Large Legal Fictions: Profiling Legal Hallucinations in Large Language Models. *Journal of Legal Analysis*. 2024;16(1):64-93. doi:10.1093/jla/laae003.
- [8] Magesh V, Surani F, Dahl M, Suzgun M, Manning CD, Ho DE. Hallucination-Free? Assessing the Reliability of Leading AI Legal Research Tools. *Journal of Empirical Legal Studies*. 2025;22(2):216-42. doi:10.1111/jels.12413.
- [9] Gehring J, Zheng K, Copet J, Mella V, Cohen T, Synnaeve G. RLEF: Grounding Code LLMs in Execution Feedback with Reinforcement Learning. In: *Proceedings of the 42nd International Conference on Machine Learning*. vol. 267 of *Proceedings of Machine Learning Research*. PMLR; 2025. p. 19034-55. Available from: <https://proceedings.mlr.press/v267/gehring25a.html>.
- [10] Blair-Stanek A, Van Durme B. LLMs Provide Unstable Answers to Legal Questions. In: *Proceedings of the Twentieth International Conference on Artificial Intelligence and Law (ICAIL 2025)*. Chicago, IL, USA: ACM; 2025. p. 425-9. doi:10.1145/3769126.3769245.
- [11] Gultekin S, Rossi Reich M, Galloni F, Lagioia F, Consiglio E, Sartor G, et al. A Two-Dimensional Evaluation Framework for Factual and Reasoning Assessment of LLMs in Legal Question Answering. In: *Legal Knowledge and Information Systems: JURIX 2025*. vol. 416 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2025. p. 86-97. doi:10.3233/FAIA251579.
- [12] Sanchi M, Novotná T. Towards Hybrid Evaluation Methodologies for Large Language Models in the Legal Domain. In: *Legal Knowledge and Information Systems: JURIX 2024*. vol. 395 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2024. p. 389-92. doi:10.3233/FAIA241279.
- [13] Garneau N, Olsen HP, Corduant A, Tarissan F. Combining Network and Text to Provide Legal Pincites. In: *Legal Knowledge and Information Systems: JURIX 2024*. vol. 395 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2024. p. 282-7. doi:10.3233/FAIA241256.
- [14] Zhang L, Savelka J, Ashley KD. Do LLMs Truly “Understand” when a Precedent Is Overruled? In: *Legal Knowledge and Information Systems: JURIX 2025*. vol. 416 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2025. p. 239-50. doi:10.3233/FAIA251592.
- [15] Nguyen HT, Satoh K. ConsRAG: Minimize LLM Hallucinations in the Legal Domain. In: *Legal Knowledge and Information Systems: JURIX 2024*. vol. 395 of *Frontiers in Artificial Intelligence and Applications*. IOS Press; 2024. p. 327-32. doi:10.3233/FAIA241263.
- [16] Trozze A, Davies T, Kleinberg B. Large language models in cryptocurrency securities cases: can a GPT model meaningfully assist lawyers? *Artificial Intelligence and Law*. 2025;33(3):691-737. doi:10.1007/s10506-024-09399-6.
- [17] Harasta J, Novotná T, Savelka J. It cannot be right if it was written by AI: on lawyers’ preferences of documents perceived as authored by an LLM vs a human. *Artificial Intelligence and Law*. 2026;34(1):153-90. doi:10.1007/s10506-024-09422-w.
- [18] Gray MA, Savelka J, Oliver WM, Ashley KD. Generating Legal Arguments with Automatically Identified Factor Magnitudes. In: *Proceedings of the Twentieth International Conference on Artificial Intelligence and Law (ICAIL 2025)*. ACM; 2025. p. 199-208. doi:10.1145/3769126.3769212.
- [19] Pires R, Malaquias Junior R, Nogueira R. Automatic Legal Writing Evaluation of LLMs. In: *Proceedings of the Twentieth International Conference on Artificial Intelligence and Law (ICAIL 2025)*. ACM; 2025. p. 420-4. doi:10.1145/3769126.3769227.

- [20] Khatri M, Sheik R, Wadhwa P, Satija G, Kumar Y, Shah RR, et al. CiteCaseLAW: Citation Worthiness Detection in Caselaw for Legal Assistive Writing. In: Legal Knowledge and Information Systems: JURIX 2023, vol. 379 of Frontiers in Artificial Intelligence and Applications. IOS Press; 2023. p. 257-62. doi:10.3233/FAIA230971.
- [21] Sadeghian A, Sundaram L, Wang DZ, Hamilton WF, Branting K, Pfeifer C. Automatic semantic edge labeling over legal citation graphs. Artificial Intelligence and Law. 2018;26(2):127-44. doi:10.1007/s10506-018-9217-1.
- [22] Dadgostari F, Guim M, Beling PA, Livermore MA, Rockmore DN. Modeling law search as prediction. Artificial Intelligence and Law. 2021;29(1):3-34. doi:10.1007/s10506-020-09261-5.
- [23] Mumford J, Florimonte F, Atkinson K, Dzehtsiarou K. Context-Aware Citation Networks: A Human-AI Dataset, Analysis, and Tool. In: Legal Knowledge and Information Systems: JURIX 2025, vol. 416 of Frontiers in Artificial Intelligence and Applications. IOS Press; 2025. p. 145-55. doi:10.3233/FAIA251584.
- [24] Spaeth HJ, Epstein L, Nelson MJ, Martin AD, Segal JA, Ruger TJ, et al.. Supreme Court Database, Version 2025 Release 01; 2025. Modern database, terms 1946 to 2024. <https://scdb.la.psu.edu/>.
- [25] Ortega JE, Joshi DD, Borkowski MP. Large-Language Memorization During the Classification of United States Supreme Court Cases; 2025. Available from: <https://arxiv.org/abs/2512.13654>. doi:10.48550/arXiv.2512.13654.
- [26] Federal Rules of Civil Procedure, Rule 1 (Scope and Purpose); 2024. "These rules govern the procedure in all civil actions and proceedings in the United States district courts". https://www.law.cornell.edu/rules/frcp/rule_1.
- [27] Cushman J, Dahl M, Lissner M. eyecite: A tool for parsing legal citations. Journal of Open Source Software. 2021;6(66):3617. Available from: <https://doi.org/10.21105/joss.03617>. doi:10.21105/joss.03617.
- [28] Liang KY, Zeger SL. Longitudinal data analysis using generalized linear models. Biometrika. 1986;73(1):13-22. doi:10.1093/biomet/73.1.13.
- [29] Norman JD, Rivera MU, Hughes DA. Reliability without Validity: A Systematic, Large-Scale Evaluation of LLM-as-a-Judge Models Across Agreement, Consistency, and Bias; 2026. Available from: <https://arxiv.org/abs/2606.19544>. doi:10.48550/arXiv.2606.19544.
- [30] Zheng L, Chiang WL, Sheng Y, Zhuang S, Wu Z, Zhuang Y, et al.. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena; 2023. NeurIPS 2023 Datasets and Benchmarks Track. Available from: <https://arxiv.org/abs/2306.05685>. doi:10.48550/arXiv.2306.05685.